- Romania has two nuclear power reactors generating about 20% of its electricity.
- The country's first reactor began operating in 1996 and its second started up in May 2007.
- Government support for nuclear energy is strong.

2 0 0

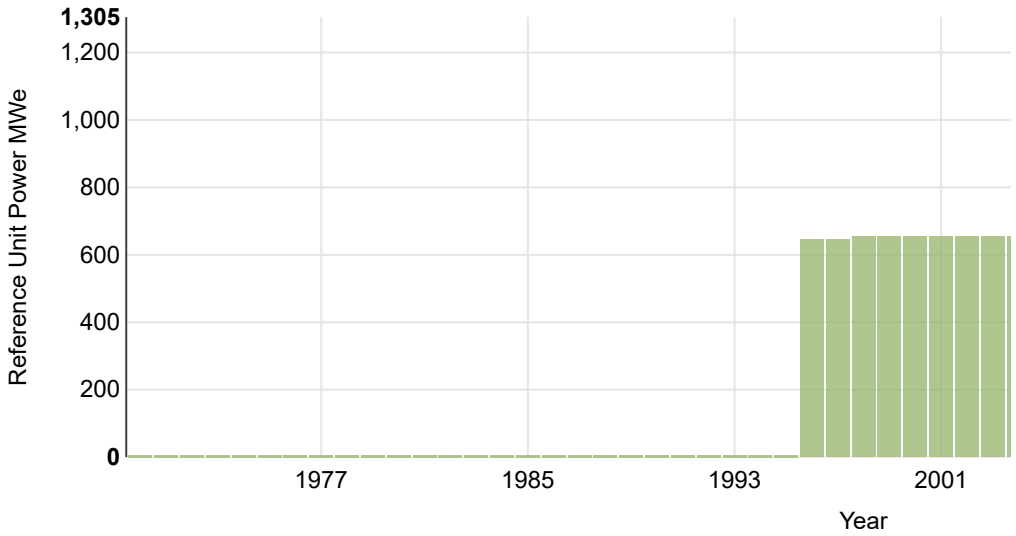
Privacy - Terms

OPERABLE REACTORS
1,300 MWe

REACTORS UNDER CONSTRUCTION
0 MWe

REACTORS SHUTDOWN
0 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 58.0 TWh

Generation mix: hydro 18.6 TWh (32%); nuclear 11.2 TWh (19%); natural gas 9.2 TWh (16%); coal 8.1 TWh (14%); wind 7.5 TWh (13%); solar 2.2 TWh (4%); oil 0.8 TWh; biofuels & waste 0.7 TWh.

Import/export balance: 3.0 TWh net import (10.1 TWh imports, 13.1 TWh exports)

Total consumption: c. 41.5 TWh

Per capita consumption: c. 2600 kWh

Source: International Energy Agency. Data for year 2023.

Nuclear power industry

Operable reactors

Reactor Name*	Model	Reactor Type	Net Capacity (MWe)	Construction Start	First Grid Connection
Cernavoda 1	CANDU 6	PHWR	650	1983-03	1996-07
Cernavoda 2	CANDU 6	PHWR	650	1983-07	2007-08

In the late 1970s a five-unit nuclear power plant was planned at Cernavoda, on the Danube River. After considering both Russian VVER-440 and Canadian Candu technology, the latter was chosen. Cernavoda was based on technology transfer from Canada (Atomic Energy Canada Limited, AECL), Italy and the USA.

Construction of the first unit started in July 1982, unit 2 in July 1983, and units 3-5 over the following four years. In 1991 work on the latter four was suspended in order to focus on unit 1, responsibility for which was handed to an AECL-Ansaldo (Canadian-Italian) consortium. Unit 1 was connected to the grid in mid-1996 and entered commercial operation in December 1996.

In 2000 the government decided that completion of Cernavoda 2 was a high priority and supplied some €60 million towards it. It was seen as the least-cost means of providing extra generating capacity for the country and construction was restarted in 2001. Further finance was raised in 2002-03, with a €382.5 million package announced by the government, including €218 million from Canada. In 2004 a €223.5 million Euratom loan was approved by the European Commission for completion of unit 2, including upgrades.

The 700 MWe (gross) unit was built by an AECL-Ansaldo-SNN management team at a total cost of €777 million.

The state nuclear power corporation Societatea Nationala Nuclearelectrica (SNN), established in 1998, operates Cernavoda. The plant provides district heating to Cernavoda township.

In September 2013 the government sold a 10% share of SNN for $85 million in an IPO, in line with commitments to the International Monetary Fund (IMF) and the EU. It also sold 1% to investment fund Fondul Proprietatea.

In September 2017 SNN decided to refurbish unit 1 from 2026. Phase I of the refurbishment project would include "activities necessary to ensure the operation of unit 1 for a period of 30 years, meaning the extension of the operating hours at nominal power in

addition to the 210,000 hours initially estimated by the design." The company noted that the overall cost would be about half that of building a new plant, and in 2019 the cost was estimated at €1.5 billion, with work scheduled over 2026-2028. In October 2020, Romania's economy minister announced that the USA would finance the refurbishment programme of unit 1, as well as the construction programme for units 3&4 (see below). Also in October, Romania and France signed a declaration of intent for a partnership to refurbish unit 1.

Refurbishing Candu 6 units involves the replacement of all calandria tubes, steam generators and instrumentation and control systems to extend the operating lifetime by at least 25 years. In January 2019, Korea Hydro & Nuclear Power (KHNP) and US engineering services firm Sargent & Lundy signed an agreement to refurbish the two units. In July 2022, Candu Energy signed the first contract for the refurbishment of unit 1, worth $49 million.

In October 2023 KHNP announced it had signed an agreement with Canada's Candu Energy and Italy's Ansaldo Nucleare to jointly carry out the second of the three phases of the Cernavoda 1 refurbishment project. The following month, SNN announced it had awarded Candu Energy and Canadian Commercial Corporation a contract to refurbish the unit worth around $750 million for engineering, technology and component procurement.

Earlier in October 2021 the Romanian government adopted the Integrated National Plan for Energy and Climate Change, submitting an updated version to the European Commission in October 2024. It confirmed plans for the construction of two Candu units at Cernavoda by 2031 (units 3&4). The plan also called for the refurbishment of the two existing units at the site, allowing each to operate for an additional 30 years.

In December 2024 Nuclearelectrica signed an EPC contract for the refurbishment works with KHNP, Candu Energy and Ansaldo Nucleare.

At the start of September 2025, civil construction works began. Later the same month, Nuclearelectrica announced it had secured a €540 million loan to support the refurbishment at unit 1, and in October an agreement was signed with Arabelle Solutions for refurbishment of the turbine generator. In January 2026, Romania filed a state aid request with the European Commission for the refurbishment of unit 1. Full refurbishment works are now expected to begin in 2027.

Reactors planned in Romania

Reactor	Type	MWe gross	Planned commercial operation
Cernavoda 3	Candu 6	720	2030
Cernavoda 4	Candu 6	720	2031
Doicești 1-6	NuScale Power Module	6x77	2033
Total (8)		1902	

Cernavoda 3&4 timeline

2002-03

In 2002 efforts got under way to resume work on Cernavoda 3 – a 700 MWe Candu 6 unit – and SNN commissioned a feasibility study from Ansaldo, AECL, and Korea Hydro and Nuclear Power (KHNP) the following year.

2004-06

In August 2004 the government advertized for companies interested in completing Cernavoda 3 through a public-private partnership arrangement. This proved impractical, and a feasibility study in March 2006 analyzed further options for both units 3&4, including that of SNN completing unit 3 itself.

2007

However, it was decided to proceed with creating a project joint venture with SNN to complete both 720 MWe units in a €2.5 billion project and then operate them. The resulting entity would be an independent power producer, with SNN providing operation and maintenance. Twelve potential investors were selected from 15 initial bidders, and in November 2007 binding offers from six companies were accepted:

- ArcelorMittal Galati (steelmaker) of Romania
- CEZ of the Czech Republic
- Electrabel of Belgium
- Enel of Italy
- Iberdrola of Spain
- RWE Power of Germany

The seven parties including SNN would spread the risk and ease the challenge of project financing.*

2008

In November an investment agreement setting up a new project company, EnergoNuclear SA, was signed between SNN, with 51% of the project, and Enel, CEZ, GDF Suez, RWE Power (each 9.15%), Iberdrola (6.2%) and ArcelorMittal Galati (6.2%).

2009

EnergoNuclear was formally established in April and embarked upon an 18-month pre-project phase. Construction cost was expected to be about €4 billion. Overall debt/equity ratio was earlier proposed to be 70/30. However, by September it was evident that SNN could not raise its share of the funds, and would contribute only 20-25%, mostly in kind – heavy water and fuel. The other participants would increase their shares accordingly, perhaps to those agreed in March 2008.

2010

In September CEZ announced its intention to sell its share, which it did to SNN by January 2011.

In February EnergoNuclear signed an agreement with AECL to assess the viability of the project and define what was required to complete units 3&4. In December the European Commission approved the project.

2011

Then in January GdF Suez, RWE and Iberdrola together gave notice of withdrawing from the project for commercial reasons. This left Nuclearelectrica (SNN) with an 84.65% share in the project and Enel holding 9.15% and ArcelorMittal Galati 6.2%.

In August SNN said that China Nuclear Power Engineering Co. (CNPEC, a subsidiary of China General Nuclear Power Group, CGN) was interested in investing in the two new Cernavoda units, and later a South Korean consortium also expressed interest. Bids were open until mid-November to partner with SNN, Enel and ArcelorMittal in EnergoNuclear, with the new investor taking about 45% of the project. Apparently no bids were received.

Three engineering contractors said they would bid for the engineering work on the two units. One was Bechtel, another a consortium of Canada's SNC Lavalin, Ansaldo Nucleare of Italy and Romanian power engineering company Elcomex IEA, and the third an AtomTechnoProm consortium of four Russian companies. All three were involved in building Cernavoda 1&2.

2012

In October the government asked the four major utilities – GdF Suez, Iberdrola, RWE and CEZ – which had withdrawn from EnergoNuclear to reconsider involvement in the Cernavoda 3&4 project.

2013

In December Enel holding and ArcelorMittal Galati also pulled out.

CGN expressed interest in investing in the project, providing non-nuclear equipment. In November two nuclear cooperation agreements were signed by SNN with CGN, one a letter of intent relating to construction of units 3&4.

2014

In May a further agreement with CGN was signed, and mid-year, the Industrial and Commercial Bank of China agreed to finance the €6.5 billion project, with CNPEC building the

two reactors. CGN would hold a major share in the project, with SNN a minor shareholder holding 49% in a new joint venture to replace or reconstitute EnergoNuclear, in which SNN then held 84.65%.

In July CNPEC signed a "binding and exclusive" cooperation agreement with Candu Energy for the construction of two more reactors at Cernavoda. The agreement complemented the November 2013 letter of intent signed by CNPEC's parent company CGN and Nuclearelectrica for investment in and development of Cernavoda 3&4. CGN itself has no experience of Candu technology.

Then in September CGN submitted its offer to build the two units and was accepted as a qualified investor. In October SNN designated CGN as the 'selected investor' for the project and the two companies signed a letter of intent to proceed.

2015

In November the two companies signed a further agreement for the development, construction, operation, and decommissioning of Cernavoda 3&4.

2016

In January the government wrote to CGN outlining major areas of support and commitment associated with the project, including electricity market reform, tariff mechanisms, electricity sales, state guarantees, financial incentive policies, and continuity of those policies. Negotiations continued for forming a joint venture company with SNN, and working out government support through long-term electricity pricing.

2019

In May CGN and SNN signed a preliminary investors agreement for the project with a view to setting up a joint venture company with 51% CGN equity in 60 days.

2020

In February the Romanian prime minister announced that the country would no longer partner with CGN for the project. In October it was announced that the USA would finance the construction of Cernavoda 3&4, as well as the refurbishment programme of unit 1.

2021

In March Nuclearelectrica said it expected to commission unit 3 by 2031, starting construction in about 2024.

The largely-new reactors would be updated versions of the Candu 6, but not the full EC6 version, since the concrete structures are already built. Unit 3 was reported to be 52% complete and 30% for unit 4, though in 2017 the reported figures were 15% and 14%. They would have operating lifetimes of 30 years with the possibility of 25-year extensions. Some 1000 tonnes of heavy water has been produced and is in storage.

2022

In December the government published a draft law establishing methods of cooperation between the state and Nuclearelectrica subsidiary Energonuclear through outlining commitments and obligations of the government needed to finance the project and enact necessary support measures, including legislation to guide the project's construction. Later that month, the Romanian government adopted the draft law.

2024
In November, Energonuclear signed the engineering, procurement and construction management contract for the completion of unit 3&4 with the FCSA joint venture of Fluor BV, Fluor Energy Transition Inc Wilmington Bucharest Branch, AtkinsRéalis (formerly SNC Lavalin), Ansaldo Nucleare, S&L Engineers Ltd and Sargent & Lundy Energie. Nuclearelectrica said that the estimated value of the contract was about €3.2 billion.
2025
In September a loan worth EUR80 million was allocated to EnergoNuclear. The loan is to cover engineering activities, structuring of project financing and European Commission approvals prior to final investment decision.

** After complex discussions, in March 2008 a draft investment agreement proposed shares of: SNN 20%, Enel 15%, Suez/Electrabel 15%, RWE Power 15%, CEZ 15%, Iberdrola 10% and steelmaker ArcelorMittal 10%. Each would provide that proportion of the financing and take the same share of the power generated. The government then decided that SNN should take 51% equity and provide funding of €1.02 billion in loans and loan guarantees. Other funds would be internal and from the partial privatization of SNN in 2011.*

Further units

SNN had been planning to complete Cernavoda unit 5 by 2020, but since this is significantly less progressed than the others, the government has been considering whether to build further nuclear capacity at other sites.

A March 2008 statement by the head of SNN said that up to four more units by 2020 at a new site were proposed. In May it was announced that Târnăveni municipality in Mureș county within the Transylvania region was favoured, with a site in the nearby Sibiu district as second choice. Three sites on the Someș River in Transylvania were also mentioned for 2400 MWe of capacity to be built after 2020. Areva had been approached to contribute to planning, with a view to a second plant being commissioned by 2030.

Small modular reactors

In March 2019, SNN signed a memorandum of understanding with NuScale power to evaluate the development, licensing and

construction small modular reactors (SMRs). In November 2021 it signed an agreement to build a 12-module NuScale SMR plant.

In May 2022 Doicești in Dâmbovița county within the Muntenia region was selected as the location for Romania's first SMR and a memorandum of understanding was signed between Nuclearelectrica and NuScale, who along with E-INFRA, would carry out engineering studies, technical analyses and licensing activities at the site. Working with NuScale, in June 2022 the US government agreed to provide $14 million of support for the engineering and design study for Romania's SMR.

At the end of December 2022 RoPower and NuScale Power signed a contract for the front-end engineering and design work for a VOYGR-6 power plant at Doicești. The same month, NuScale announced it had completed the standard generic plant design for the VOYGR plant that would serve as a starting point for deploying site-specific designs.

In May 2023 the USA, along with multinational public-private partners from Japan, South Korea and the United Arab Emirates, announced funding of up to $275 million for the deployment of the SMR plant. A control room simulator for the plant was opened earlier that month.

In October 2024 the US Export-Import Bank approved a final commitment for a $98 million loan for pre-project services.

In February 2026, Nuclearelectrica shareholders approved the final investment decision for the NuScale VOYGR-6 project at Doicești, with the first module targeted for commercial operation in 2033.

Fuel cycle

Fuel

Up to 1963 some 20,000 tonnes of uranium was supplied to the Soviet Union from mines at Baita (Bihor area in the northwest) and at Banat in the southwest. The Baita mine produced very small output through to 2008 from decommissioning. Considerable rehabilitation of early mine sites has been undertaken by the National Commission for Nuclear Activities Control (CNCAN).

The state-owned Compania Nationala a Uraniului (CNU) had been producing about 50 tU per year from 3100 tU known resources at

Crucea-Botusana in the north, with government subsidy. These mines were commissioned in 1983 and 1985 but are now depleted. Production ceased in 2017. Ore treatment is 350 km away in the Uzina mill at Feldioara in the centre of the country, using pressure alkaline leaching. This plant was commissioned in 1978. CNU was planning to develop the small Tulghes-Grinties deposit (discovered in 1960s) in the East Carpathian mountains about 100 km south of Crucea-Botusana at a cost of €91 million. Another deposit is in the Highis-Drocea Mountains in the west.

In 2014 it was announced that CNU would merge with SNN, but this did not occur and SNN sources lower-priced uranium from abroad. Since 2016 CNU has been receiving state aid. In November 2021 SNN was cleared to take over CNU and the Feldioara plant, completing the acquisition of the Feldioara assets in December 2022. In October 2025, the Feldioara plant won a contract to supply uranium dioxide to Argentina, marking Romania's first nuclear fuel export.

Cernavoda 1 has been using 105 tonnes of natural uranium oxide fuel per year, which is fabricated by the SNN subsidiary Fabrica de Combustibil Nuclear (FCN) Pitesti fuel plant in the south of the country. This was qualified by AECL in 1994 and remains the only such plant producing Candu fuel outside Canada.

Prior to 1990 some 31,000 fuel bundles were made, but these were assessed as unusable so were dismantled and refabricated. This material comprised half of the fuel used to early 2007.

At the end of 2003 FCN started making slightly heavier fuel bundles, and in preparation for unit 2 commissioning, its capacity was doubled to 46 fuel bundles per day.

Heavy water was produced by the Romanian Nuclear Activities Authority (RAAN) in the southwest of the country, near Drobeta-Turnu Severin. The plant is now being decommissioned.

Waste management

Used fuel is stored at the reactors for six to ten years. It is then transferred to a dry storage facility (DICA) at Cernavoda based on the Macstor system designed by AECL, where it will remain for about 50 years as the responsibility of SNN. The first module was commissioned in 2003.

At Saligny near Cernavoda, a low- and intermediate-level waste repository (DFDSMA) for material from operation and

decommissioning of Cernavoda is under construction by the Nuclear and Radioactive Waste Agency (ANDR), which is responsible for final disposal. This will be a surface repository consisting of concrete cells, with containers of cement-conditioned waste, covered by a concrete cap and finally by a multilayer impervious cover. SNN applied for a licence in 1998, and in 2008 a partial site licence was issued by CNCAN. The repository is expected to cost €40 million for the first of eight stages. It will be funded from the €2/MWh fee paid by SNN, which is also for reactor decommissioning.

Preliminary investigations have been under way since 1992 regarding a deep geological repository, and six geological formations are considered potential, particularly the green schists of Central Dobrogea which are being assessed by NAGRA from Switzerland for ANDR.

The national repository for industrial low-level radioactive waste (DNDR) has operated since 1985 at Baita Bihor in two galleries of the former Baita uranium mine. Its capacity is 21,000 standard 220-litre drums. Under Romanian law, waste rock and mine tailings are considered to be radioactive waste.

Research and development

The Romanian Nuclear Activities Authority (RAAN) operated the Triga research reactor and undertook R&D on safety, nuclear fuel, radiation protection, reactor systems, and radioactive waste management. It also operated the heavy water plant. In 2013 it became the Technologies for Nuclear Energy State Owned Company (*Regia Autonomă Tehnologii pentru Energia Nucleară*, RATEN), comprising two divisions: the Institute for Nuclear Research Pitesti (ICN) and the Center of Technology and Engineering for Nuclear Projects (CITON).

ICN was founded in 1971 under the State Committee for Nuclear Energy, and eventually became the R&D subsidiary of RATEN in 2013. It is responsible for Romanian involvement in the EU's Advanced Lead Fast Reactor European Demonstrator (ALFRED) project and will host it.

CITON undertakes project engineering services, design and construction management, maintaining links with Candu Energy, Ansaldo, etc. It was the general design authority (1979-1991) and design and engineering main contractor (since 1992) for the Cernavoda plant. It is also the general design and engineering

authority for the decommissioning of the VVR-S research reactor and the heavy water plant.

A 14 MWt Triga research reactor is operated by ICN at Pitesti, and a 2 MW Russian VVR-S unit is being decommissioned at Bucharest-Magurele.

A consortium between Italy's National Agency for New Technologies, Energy and the Environment (ENEA) and Ansaldo Nucleare, as well as Romania's Institute for Nuclear Research Pitesti (*Institutul de Cercetari Nucleare Pitesti*, ICN), was set up in December 2013 for the construction of the Advanced Lead Fast Reactor European Demonstrator (ALFRED) in Romania. Named the Fostering Alfred Construction (Falcon) consortium, the group will be expanded through the participation of further European organizations. The total cost of the project is put at some €1.0 billion. ALFRED will be built at ICN's facility in Mioveni, near Pitesti in southern Romania.

The reactor is being developed through the European Sustainable Nuclear Industrial Initiative (ESNII), which brings together industry and research partners in the development of Generation IV fast neutron reactor technology, as part of the EU's Strategic Energy Technology Plan (SET-Plan). ESNII was set up under the umbrella of the Sustainable Nuclear Energy Technology Platform (SNETP), formed in 2007 and bringing together more than 90 stakeholders involved in nuclear fission.

The 300 MWt ALFRED project is seen as a precursor to an industrial demonstration unit of about 600 MWe. The lead-cooled reactor will employ mixed oxide (MOX) fuel with about 17% plutonium, and will operate at temperatures of around 550°C. It features passive safety systems.

In November 2021 a consortium including Ansaldo Nucleare and Romania's Reinvent Energy was awarded a €20 million contract for the design, procurement, installation, and commissioning of the Advanced Thermo-Hydraulics Experiment for Nuclear Application (ATHENA) facility at the ICN research centre near Pitesti. The 2.21 MWt pool-type reactor with 880 tonnes of liquid lead would support the development of the ALFRED project.

In March 2025, the €112 million ‘4ALFRED’ project was announced, covering four experimental facilities to support development of the ALFRED reactor.

Regulation

The National Commission for Nuclear Activities Control (CNCAN) is the regulator which was set up under the Nuclear Act 1996 to ensure safety and to licence nuclear sites and operations including waste storage and disposal. It is also responsible for safeguards and other international liaison, ensuring conformity with International Atomic Energy Agency (IAEA) standards, as well as radiation protection.

The Nuclear and Radioactive Waste Agency (ANDR) was established in 2009 by merging two previous entities.

Non-proliferation

Romania is a party to the Nuclear Non-Proliferation Treaty (NPT) since 1970 as a non-nuclear weapons state. The Additional Protocol in relation to its safeguards agreements with the IAEA came into force in 2000. It is a member of the Nuclear Suppliers Group.

Notes & references

General sources

National Commission for Nuclear Activities Control (CNCAN)